

Unit 3 – Section 3: Decision-Making Techniques

One-Page Student Summary Sheet

Key Techniques and Definitions

Technique	Definition	Real-World Example
Quantitative Sales Forecasting (QSF)	Uses historical data (e.g., moving averages, correlation) to predict future sales.	Tesco using 3-year moving averages to plan stock levels.
Investment Appraisal	Evaluates financial viability of projects using Payback, ARR, and NPV.	Apple assessing ROI before launching a new product.
Payback Period	Time taken to recover initial investment from cash inflows.	A retail chain deciding between two refurbishment options.
Average Rate of Return (ARR)	Average annual profit as a percentage of initial investment.	Tesla assessing new battery factory profitability.
Net Present Value (NPV)	Present value of cash inflows minus outflows, considering discount rates.	Amazon using NPV to decide on new warehouse expansion.
Decision Trees	Diagrams showing options, probabilities, and expected monetary values (EMVs).	Toyota using EMVs to choose between new car models.
Critical Path Analysis (CPA)	Network diagram showing the shortest possible project completion time.	Heathrow Airport planning runway construction schedule.
Float Time	Extra time an activity can take without delaying the project.	Marketing campaign steps with flexible deadlines.
Contribution	Selling price minus variable cost per unit; used in short-	Greggs deciding whether to accept a low-margin catering

	term decisions.	order.
Special Orders	One-off orders outside normal business at lower prices or margins.	A bakery supplying hotels for a one-time event.

Command Words & AO Focus

Mark	Command Words	AO Skills	Tips for Students
4 marks	Define / Calculate / Explain	AO1 + AO2	Write formula, show full working, and give one example.
8 marks	Assess	AO2 + AO3	Two balanced arguments with context and link to results.
12 marks	Discuss	AO3 + AO4	Develop both sides of argument, add “depends on” evaluation.
20 marks	Evaluate	AO1–4 (all)	Combine analysis, numerical results, and strong overall judgement.

High-Frequency Question Themes

- NPV vs ARR – most accurate method debate.
- QSF – reliability and external factors.
- Decision Trees – quantitative vs intuitive decision-making.
- CPA – efficiency and project management.
- Contribution – short-term vs long-term use in pricing.

Revision Checklist

- ✓ Can I calculate ARR, NPV, payback, EMV, and float accurately?
- ✓ Can I interpret and explain decision trees and CPA diagrams?
- ✓ Can I balance quantitative data with qualitative judgement?

✓ Can I evaluate whether these techniques truly improve decisions?